SAIRAM ANBALAGAN

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years of experience building and deploying ML solutions across commercial and financial services. Specialized in building NLP-powered recommendation systems, conversational AI and CI/CD pipelines with a strong focus on building scalable and secure AI/ML platform in AWS environment. Passionate in leveraging agentic skills to assist with protoyping innovative use cases and creating workflow automation.

CORE SKILLS

Languages: Python, R, SQL, Bash, YAML

Frameworks: Pandas, NumPy, TensorFlow, PyTorch, LangChain, Hugging Face, FastAPI, MLflow

Tools & Platforms: GitHub Actions, Kubernetes, Docker, JupyterLab, Terraform, Tableau

AI Agents: LLamaindex, LangGraph, smolagents

Cloud & Infrastructure:S3, ECS, Sagemaker, Bedrock, CloudFormation, CloudWatch, Agentcore

EDUCATION

University College Dublin, Ireland **2020 - 2021**

Msc. Data and Computational Science

GPA: 3.79 (1:1 Honors)

Coursework: Neural Networks, Deep Learning, NLP, Python, Statistics

Panimalar Institute of Technology, India **2014 - 2018**

BE Electronics and Communication Engineering

GPA: 7.8

EXPERIENCE

Tata Consultancy Services **Dublin, Ireland**

Machine Learning Engineer **Nov 2021 - May 2025**

AI/ML Platform management - Prudential Insurance

• Owned and managed Florence AI/ML platform, ensuring end-to-end operational reliability and troubleshooted issues across AI services consumed by 20+ data scientists.
• Configured and enabled new AI capabilities such as Bedrock, AgentCore to drive GenAI initiatives.
• Built and managed lifecycle configuration pipelines for LLM model persistence and streaming optimization.
• Built API regression test suite to perform healthchecks of GenAI models in API gateway.
• Optimized Cloud infrastructure expenditure by $300k by developing cost optimization pipeline to track cost centers and free up unused resources.
• Assisted Data Scientists to deploy API in Sagemaker endpoints by performing unit testing, canary testing, model monitoring and observability.

• **ML model development - Prudential Insurance**

RAG-based insurance chatbot deployment:

• Developed conversational AI in AWS Bedrock to handle insurance policy and claims queries by policy owners across 15 languages which assisted in reducing 500+ hours of manual query handling.
• Implemented document ingestion pipeline that leverages OCR and PDF processing tools to enable efficient semantic searches in pinecone database. Added guardrails to deny deviating topics and handling harmful content.
• Created evaluation jobs that sends prompt datasets to curated Judge model to provide automated reports. Ensured high output quality by consistently testing random sample of response and providing feedback on highly sensitive and edge case scenerios.
• Leveraged cache checkpoints to reduce inference response latency and input token costs by 20%.
• Performed initial A/B testing on 10000 users and iteratively rolled out to 200k active monthly users while proactively fixing deployment issues.

Predictive Auto-Tagging model to refine enterprise data pipelines:

• Deployed a monthly batch job pipeline that intakes data from various customer and policy sources and leverages LLM-based entity recognition with similarity scores to standardize inconsistent identifiers and store output in S3 bucket.
• Leveraged logging and observability metrics for triggering alerts and obtaining data quality metrics.
• The automation effort enabled in reducing manual tagging effort by 200+ hours per quarter and accelerated data-onboarding workflows for the enterprise data platform teams.

NLP Recommendation Engine - Circler

• Served as the primary AI/ML engineer, owning the end-to-end development of the recommendation engine and reporting directly to leadership.
• Leveraged ALS scoring to learn latent relationship between cluster of product and customer categories and new users with similar user features are recommended with the product with closest similarity.
• Developed an NLP-powered system analyzing customer reviews with LLMs to extract purchase intent, sentiment and style preferences and the score generated from weighted combination of collaborative filtering and NLP signals is used to create final recommendation for the customer.
• Implemented data preprocessing pipelines to clean product and customer data, storing standardized features in a feature store for reusable inputs across ML models and enabling scalable model retraining.
• The real time recommendation endpoint is served via FastAPI optimized for low latency inference. The scoring system assisted with 50% increase in clickthrough rate which translated into 20% increase in product purchase.

Infosys **Chennai, India**

Systems Engineer **May 2018 - Aug 2019**

• Customer risk predictive system: Developed hybrid risk scoring system by leveraging Random Forest to improve pricing accuracy and customer segmentation for the Full Service Dental product. The recommendation system assisted in lowering claims payout by 16% and increased revenue by $150k.

• Built BI dashboards in Tableau for Total Long Term Care product to understand claims and customer patterns to identify risk drivers and pricing gaps; enabled faster, data driven decisions that boosted quarterly sales by 30%.

• Monitored and debugged critical JCL job streams on IBM z/OS, diagnosing root causes of data and logic failures; implemented fixes that reduced recurring abends by 40% and ensured on-time batch completion across daily operations.